1. Identity

Document	STAR-PKG-STAR-MOON-EN · v1.3 · 2026-10-16
Package	star-moon
Version	0.1.0
Capability	star.orbit.moon
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_moon

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_moon-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_moon-0.1.0-py3-none-any.whl	a50cb2e2e106001163cd963b32cfdf612904dbf06417c388f081dd2f520e222d
star_moon.py	ce41f15b68f4841d07880ec30babf3cb98a4e8c80bdb0ec867b6f6a0872cd769
star_moon-0.1.0-source.zip	8d4319952b295c78dc288a8e37acf8664d62df20eb4f6e29f261c588b489b582

5. What it does, and its limits

Geocentric Moon position for 1950-2050 from the Astronomical Almanac series; measured within 0.36 deg and 1200 km of JPL DE440 over 2002 dates, with the same envelope against astropy and PyEphem.

Not an ephemeris: 0.36 deg is about 2300 km at the Moon's distance. Good for eclipse and illumination geometry, third-body direction and visibility planning; not for navigation, occultation timing or tide-accurate work.

- `moon_vector_km(jd_day, jd_frac=0)`, `moon_ra_dec_deg`, `moon_distance_km` for Julian dates (TDB) from 1950-01-01 to 2050-01-01; outside, `ValueError`. Reproduces Vallado Example 5-3 within 0.4 m.
- Measured accuracy of the series (2002 dates, three ephemerides): direction within 0.36 deg of JPL DE440 (95 % of the dates within 0.20 deg), distance within 1200 km (288 km on average). astropy and PyEphem give the same envelope.
- Frame: mean equator and equinox of DATE, like `star_sun`; rotate before mixing with J2000/GCRF vectors.
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, a date outside 1950-2050.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_moon_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-moon}: Where the Moon is, to about 0.4 degree (0.2 degree for 95 % of
dates)},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_moon/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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Related: space activities and export control · acceptable use · evidence and limits · licence texts